

Example. Lets write $n^i = p^i/|p|$. Recalling that for both representations, the Lorentz generators can be written as $M^{ij} = \epsilon^{ijk}\sigma^k/2$:

$$H = S_i n^i = \frac{1}{2}\epsilon_{ijk}M^{jk}n^i = \frac{1}{2}\epsilon_{ijk}\left(\frac{1}{2}\epsilon^{jkl}\sigma^l\right)n^i = \frac{1}{4}(2\delta_i^l\sigma^l)n^i = \frac{1}{2}\sigma_i n^i$$

In vector notation, we can write $\vec{\sigma} \cdot \vec{p}/2$ where the components of $\vec{\sigma}$ are the Pauli matrices.

(b) Recalling that the Pauli matrices are orthogonal and idempotent, we have

$$H^2 = \frac{1}{4}(\sin^2 \theta \cos^2 \phi \sigma_1^2 + \sin^2 \theta \sin^2 \phi \sigma_2^2 + \cos^2 \theta \sigma_3^2) = \frac{1}{4}(\sin^2 \theta \cos^2 \phi + \sin^2 \theta \sin^2 \phi + \cos^2 \theta) = \frac{1}{4}$$

Therefore, the eigenvalues of H are $\pm \frac{1}{2}$. Writing the matrix as

$$H = \frac{1}{2} \begin{pmatrix} \cos \theta & e^{-i\phi} \sin \theta \\ e^{i\phi} \sin \theta & -\cos \theta \end{pmatrix}$$

We can find the vectors with eigenvalues $+1/2$ and $-1/2$ as

$$\xi^+(\vec{p}) = \begin{pmatrix} \cos(\theta/2) \\ e^{i\phi} \sin(\theta/2) \end{pmatrix} \quad \& \quad \xi^-(\vec{p}) = \begin{pmatrix} -e^{-i\phi} \sin(\theta/2) \\ \cos(\theta/2) \end{pmatrix}$$